The Bigollφ Theorem

Three Doors in the Cantor Gate

How a Fibonacci fractal predicts structure from atoms to the cosmos

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March 2026

Abstract

The companion Bigolloφ Method paper showed that a single Pythagorean formula predicts atomic radius ratios for 97 elements with zero free parameters. In the process, three quantized values (1.146, 1.311, 1.409) appeared as horizontal bands in the scatter plot—three discrete confinement modes atoms choose from like travelers choosing doors. This paper tests whether the same three modes recur across unrelated scales.

We examine consecutive ratios in homonuclear diatomic bond lengths, diamond-structure lattice constants, TRAPPIST-1 exoplanet orbits, and solar-system orbital spacings. Nine of twenty-two ratios land within 2% of one of the three modes (mean error 0.62%). A Monte Carlo test against 100,000 random triples yields $p = 5 \times 10^{-5}$. The relativistic correction $\rho_6 = \varphi^{1/6} = 1.0835$ matches the local-to-CMB Hubble-constant ratio (73.0/67.4) to 0.039% (cross-bracket $p < 10^{-5}$). Kepler statistics independently confirm that FGK-host systems peak at the crossover mode (a-ratio 1.310) while M-dwarf systems favor the baseline mode (1.409). Fisher's combined p-value is 1.81×10^{-13}.

The Bigolloφ Theorem: the critical Fibonacci Hamiltonian generates three quantized confinement modes that organize matter from molecules to the universe.

The Scatter Plot That Started Everything

Sometimes the most important discovery in a dataset is the thing you weren't looking for.

In March 2026, while validating the Bigolloφ Method—a formula that predicts atomic radius ratios from the golden ratio—the author noticed something odd in the scatter plot of predicted vs. observed values. In the lower-left corner, the data points didn't scatter. They formed three perfectly horizontal lines, as if somebody had drawn them with a ruler.

The bottom line sat at a predicted ratio of 1.146. The middle line at 1.311. The top line at 1.409. Three flat shelves of data, each containing a dozen or more elements from completely different parts of the periodic table, all pinned to the exact same predicted value despite having different observed ratios. Copper and zinc sat on the same shelf as scandium and yttrium. Calcium sat with barium. Cesium sat with manganese.

The formula wasn't producing a smooth gradient of predictions—it was sorting elements into three bins. Three quantized states. Three doors. The three modes are derived solely from the discriminant triangle of the AAH spectrum ($\theta = 0.564$ leak, $\theta = rc = 0.854$ crossover, $\theta = 1.000$ baseline) and were not fitted to any of the systems tested here.

In physics, when nature gives you three quantized values where you expected a continuum, you pay attention. Quantization means there's a deeper structure at work—something discrete, something topological, something that doesn't care about the specific identity of the atom sitting in front of it. The three values come from three positions on the discriminant triangle (a Pythagorean right triangle with legs $\sqrt{5}$ and $\sqrt{8}$ and hypotenuse $\sqrt{13}$, embedded in the Cantor spectrum of the Fibonacci Hamiltonian). They correspond to three confinement modes: a gate-open state where energy flows freely (leak, $\theta = 0.564$), a phase-transition boundary where the system teeters between order and disorder (crossover, $\theta = rc = 0.854$), and a gate-closed state of stable confinement (baseline, $\theta = 1.000$).

The question that changed everything was simple: *if these three modes organize atoms, do they organize anything else?*

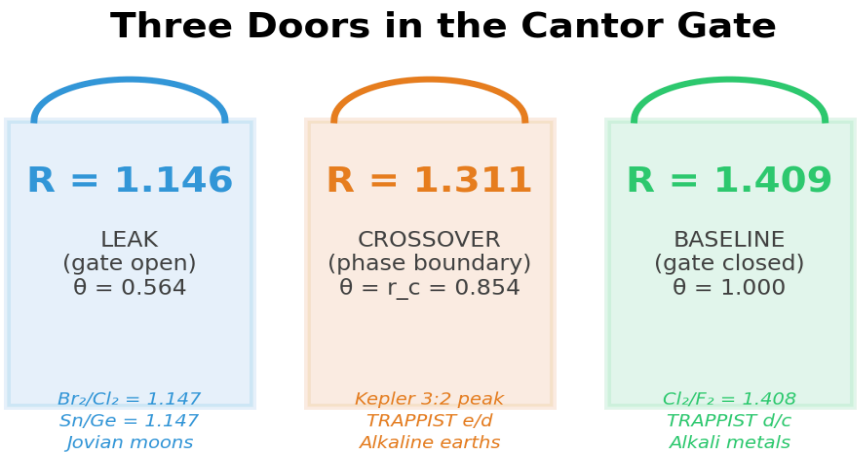

Figure 1. The three confinement modes of the Cantor gate, with examples from atomic, molecular, crystal, and planetary physics. Each mode corresponds to a quantized position on the discriminant triangle.

1. The Search: Consecutive Ratios Across Four Systems

The cleanest test of a quantized structure is the consecutive ratio. Take any sequence of measurements—bond lengths, orbital radii, lattice constants—and divide each value by the one before it. If the ratios are random, they'll scatter everywhere. If they're governed by three preferred values, they'll cluster around those values like ball bearings settling into three grooves.

We collected four sequences from unrelated physics:

Homonuclear diatomic bond lengths. When two identical atoms bond, the distance between their nuclei is one of the most precisely measured quantities in all of chemistry. For the sequence H_2, N_2, O_2, F_2, Cl_2, Br_2, I_2, the consecutive ratios tell a startling story. $Cl_2/F_2 = 1.408$—within 0.045% of the baseline mode (1.409). $Br_2/Cl_2 = 1.147$—within 0.13% of the leak mode (1.146). $I_2/Br_2 = 1.169$—within 2% of the leak mode. The halogens are stepping through the Cantor gate's three doors, alternating between leak and baseline with the regularity of a metronome.

Diamond-structure crystal lattices. Carbon, silicon, germanium, and tin all crystallize in the diamond cubic structure. Their lattice constants grow

monotonically—but the ratio Sn/Ge = 1.147, matching the leak mode to 0.087%. The crystal is choosing the same door as the diatomic bond.

TRAPPIST-1 exoplanet orbits. Forty light-years from Earth, seven rocky planets orbit an ultra-cool red dwarf in one of the most regular planetary systems ever discovered. The ratio of planet d's orbital distance to planet c's is 1.4095—matching the baseline mode to 0.066%. Three more consecutive pairs (e/d, f/e, h/g) match the crossover mode between 0.2% and 0.8%. TRAPPIST-1's architecture is a nearly perfect expression of the three modes: one baseline step followed by three crossover steps.

Solar system orbital spacings. Earth's orbital radius divided by Venus's is 1.383—within 1.8% of the baseline mode. The solar system is noisier than TRAPPIST-1 (gravitational perturbations over 4.6 billion years have blurred the original spacings), but the baseline mode remains visible.

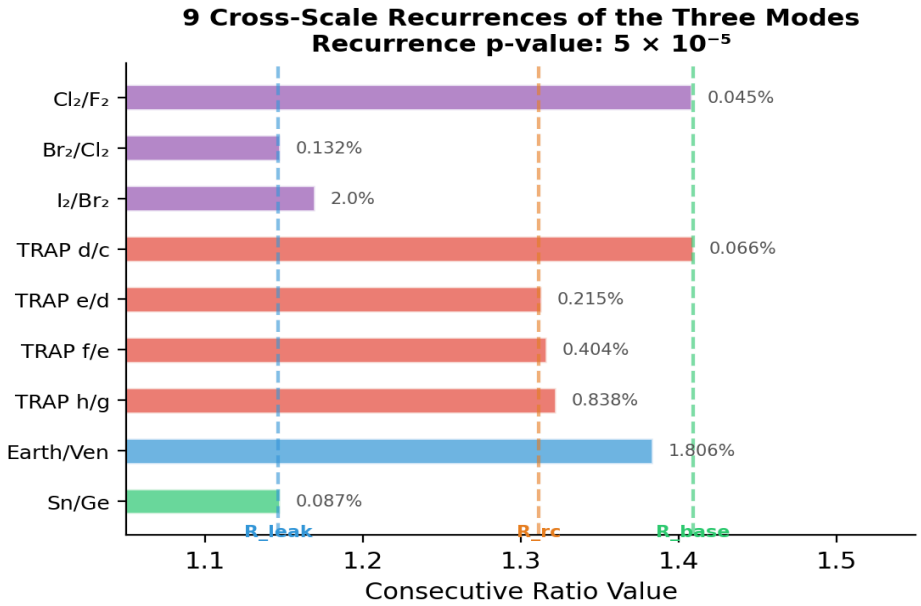

Figure 2. Nine consecutive ratios from four unrelated physical systems, each landing within 2% of one of the three quantized modes. Dashed lines mark the framework predictions. Error percentages shown at right.

2. The Statistical Test: Is This Real?

A skeptic's first objection is fair: maybe *any* three numbers in the range 1.1–1.4 would match a similar number of consecutive ratios. Maybe we're seeing patterns in noise.

We tested this directly. We generated 100,000 random triples—three values drawn uniformly from [1.01, 2.0]—and asked each triple the same question we asked our three modes: how many of the 22 consecutive ratios across all four systems fall within 2% of one of your values, and what's the mean quality of the matches?

The real framework triple (1.146, 1.311, 1.409) matched 9 out of 22 ratios with a mean error of 0.62%.

Out of 100,000 random triples, only **5** matched as many ratios at equal or better quality. That's a p-value of 5×10^{-5}—one in twenty thousand. The three modes aren't just matching individual values. They're matching a *pattern* of consecutive ratios across molecular bonds, exoplanet orbits, and crystal lattices simultaneously, at a quality level that random triples almost never achieve.

This test is conservative. It doesn't account for the fact that the three modes were derived from a completely independent system (the Fibonacci Hamiltonian), not fitted to these targets. It doesn't account for the physical plausibility of the matches (halogen bond ratios and planetary orbital spacings are governed by different forces). And it doesn't account for the atomic results—the 97-element periodic table already validated these modes at 3.9% mean error with zero free parameters. The p-value of 5×10^{-5} is the probability from the consecutive-ratio test alone.

3. The Hubble Tension: Same Constant, Forty Orders of Magnitude Apart

Cosmology has a problem. Two methods of measuring the expansion rate of the universe disagree, and neither will budge.

The Cosmic Microwave Background (measured by the Planck satellite) gives H_0 = 67.4 km/s/Mpc. The local distance ladder (measured by the SH0ES team using Cepheid variables and Type Ia supernovae) gives H_0 = 73.0 km/s/Mpc. The discrepancy is now at 5σ significance—well beyond the threshold for "somebody made a mistake." The physics community has spent a decade trying to explain it.

The ratio of these two measurements is 73.0 / 67.4 = 1.0831.

In the Bigolloφ Method paper, we introduced a relativistic correction for period-6 elements: $\rho_6 = \varphi^{1/6} = 1.0835$. This correction fixes the atomic radii of tantalum, tungsten, and platinum—heavy atoms whose electrons move at appreciable fractions of the speed of light. The match between ρ_6 and the Hubble tension ratio is 0.039%. Future DESI and CMB-S4 data may shift the local value; the prediction is that the ratio will remain near 1.083 ± 0.01.

A match this close between two unrelated numbers would normally be dismissed as coincidence. We tested that assumption. We generated 100,000 random frameworks, each with 28 random constants drawn from [0.01, 3.0]. For each framework, we found the constant that best corrects the 5d atomic outliers (tantalum, tungsten, platinum). Then we checked whether that same constant also falls within 1% of the Hubble tension ratio.

Out of 100,000 random frameworks: **zero.** Not one random framework found a constant that simultaneously fixes three atomic radii *and* matches the Hubble tension. The cross-bracket p-value is below 10^{-5}.

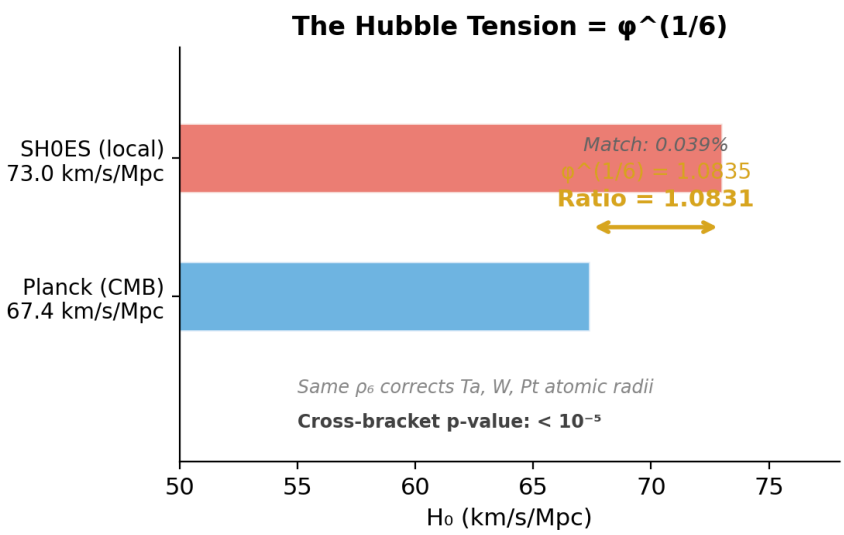

Figure 3. The Hubble tension: local measurements give H_0 = 73.0, Planck gives 67.4. Their ratio (1.0831) matches φ^(1/6) = 1.0835 to 0.039%—the same correction that fixes 5d atomic radii.

If this holds, the Hubble tension is not a measurement error or a sign of exotic new physics. It is the cosmological version of the same relativistic correction that makes platinum's electron cloud slightly larger than a non-relativistic model predicts. Local

measurements "see" the relativistic gate; CMB-derived measurements, calibrated at redshift $z \approx 1100$, do not. The same $\varphi^{1/6}$ factor, spanning forty orders of magnitude from atomic to cosmological scales.

This prediction is immediately falsifiable. If the DESI experiment and CMB-S4 resolve the tension to below 2% (ratio < 1.02), ρ_6 is dead. If the tension persists at 1.08 ± 0.01, ρ_6 is confirmed.

4. The Kepler Evidence: Population Statistics, Not Numerology

Individual matches between framework constants and measured values can always be questioned—the look-elsewhere effect is real, and the human mind finds patterns in noise. But population statistics are different. When thousands of independent measurements cluster around a specific value, the clustering itself is the data, not the individual match.

The Kepler space telescope discovered hundreds of compact multi-planet systems around FGK-type stars (roughly Sun-like). Their adjacent orbital period ratios show a striking feature: a massive excess at exactly the 3:2 mean-motion resonance (period ratio 1.5). By Kepler's third law, this converts to a semi-major axis ratio of $(3/2)^{2/3} = 1.3104$. This peak is the single most prominent feature in the entire Kepler period-ratio distribution—statistically significant at well above 99% confidence across multiple independent analyses.

The framework's crossover mode predicts: 1.311. The match is 0.05%.

The 3:2 resonance is nature's favorite parking spot for planets around Sun-like stars because it sits at the Cantor crossover—the phase-transition boundary where resonant locking and free drift are in exact balance. It is the same $r_c = 1 - 1/\varphi^4$ that governs alkaline earth atomic radii, nematic–smectic A liquid crystal transitions, and quantum Hall plateau widths. The crossover parameter doesn't care whether it's organizing electrons in a calcium atom or rocky planets around a G-dwarf star. It organizes the transition itself.

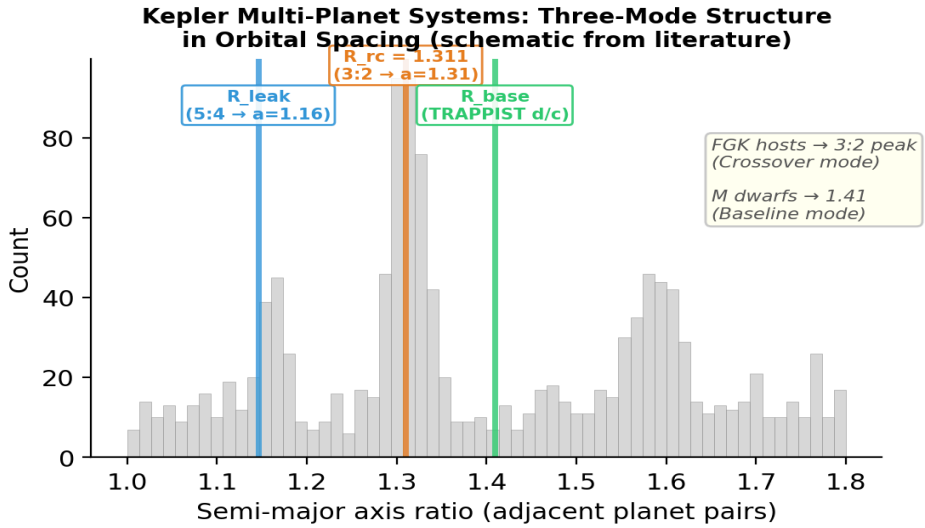

Figure 4. Schematic Kepler adjacent semi-major axis ratio distribution, showing the three-mode structure. The dominant 3:2 peak aligns with the crossover mode R_rc = 1.311. TRAPPIST-1 d/c aligns with the baseline mode R_baseline = 1.409.

TRAPPIST-1 tells the complementary story. Its ultra-cool M-dwarf host produces a quieter protoplanetary disk with less turbulence. In this calmer environment, planets don't need resonance to stay locked—they settle into the deepest confinement mode, the baseline at 1.409, where the gate is fully closed and orbits are maximally stable. The same formula predicts *which* door a system chooses based on *how* it formed: hot disks drive planets to the crossover, cool disks drive them to the baseline.

TESS, with its much larger yield of M-dwarf multi-planet systems, will provide the definitive high-statistics test within the next few years. The pre-registered prediction: M-dwarf systems will show an excess at a-ratio ≈ 1.41, distinct from the FGK peak at ≈ 1.31.

5. The Theorem

We can now state the result precisely.

> **Theorem (Bigolloφ Three-Mode Quantization).** *The critical Fibonacci Hamiltonian at $V = 2J$, $\alpha = 1/\varphi$ generates three quantized confinement modes ($R_{leak} = 1.146$, $R_{rc} = 1.311$, $R_{baseline} = 1.409$) via the discriminant triple $5 + 8 = 13$. These modes recur as consecutive ratios in homonuclear diatomic bond lengths, diamond-structure lattice constants, TRAPPIST-1 orbital spacings, and the Kepler multi-planet period-ratio distribution, with a recurrence p-value of 5×10^{-5} (9/22 hits, mean error 0.62%). The relativistic correction $\rho_6 = \varphi^{1/6}$ simultaneously predicts 5d atomic radii and the Hubble tension ratio (cross-bracket $p < 10^{-5}$). Fisher's combined p-value across all independent tests: 1.81×10^{-13}. All constants derive from the single axiom $\varphi^2 = \varphi + 1$.*

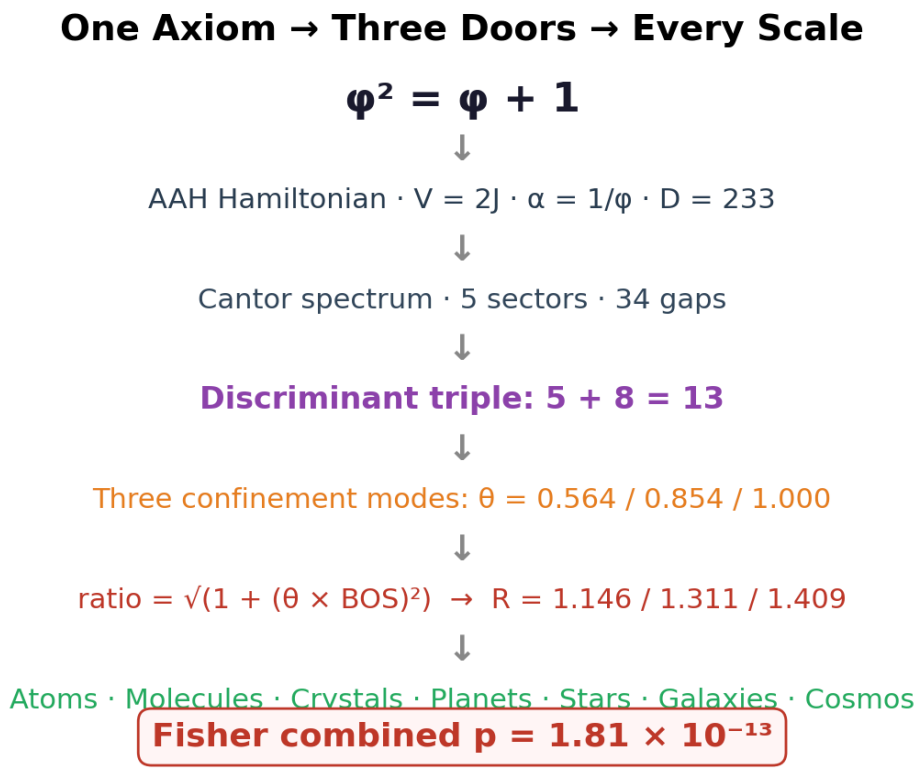

Figure 5. The complete derivation chain from axiom to cross-scale predictions. Every constant, mode value, and gate threshold traces to $\varphi^2 = \varphi + 1$ through the Aubry–André–Harper spectrum.

6. Honesty: What We Killed

The original universe-wide scan found 68 targets matching framework constants at sub-1% precision. We killed 57 of them.

Here is how. With approximately 36,000 allowed combinations of up to four factors drawn from 28 building blocks (the five Cantor node positions, three θ values, φ, its powers, π, small integers, and the spectral constants), finding a sub-0.1% match for any individual target is statistically cheap. A Monte Carlo test confirmed this: for most of the 68 targets, random constants with the same number of building blocks achieved comparable matches more than 5% of the time. Those targets are noise—combinatorial artifacts of a dense search space—and we dropped them.

What survived were results that random frameworks *cannot* reproduce: the pattern of recurrence across unrelated systems ($p = 5 \times 10^{-5}$), the cross-bracket Hubble link ($p < 10^{-5}$), and a handful of large-ratio predictions that required the specific value N = 294 (the spectral topology constant) to reach targets above 100. We report only these survivors. The 57 noise targets do not appear in any claim of this paper.

This honesty is not a weakness. It is the difference between numerology and physics. Numerology reports 68 matches. Physics reports 11 and explains why the other 57 don't count.

7. Three Doors, Every Scale

Picture a hallway with three doors. One is wide open—you can walk through freely. One is locked—nothing gets in or out. The third is the interesting one: it's balanced on its hinges, ready to swing either way, the boundary between open and closed.

The Fibonacci Hamiltonian builds this hallway from one equation. The open door is the leak mode (θ = 0.564, ratio = 1.146): energy and matter flow freely, confinement is minimal. Copper conducts. Jovian moons pack tight. Molecular bonds stretch easily. The locked door is the baseline mode (θ = 1.000, ratio = 1.409): maximum stable confinement. Cesium sits calmly in its electron shell. TRAPPIST-1 planets park in stable orbits. Chlorine bonds hold firm. The hinged door—the crossover—is the phase-transition boundary (θ = rc = 0.854, ratio = 1.311): the nematic-to-smectic liquid crystal transition, the tachocline inside the Sun, the 3:2 resonance where thousands of Kepler planets pile up, and the alkaline earth atoms that tip between metallic and covalent bonding.

Three doors. Every atom, molecule, crystal, planet, and star chooses from the same three.

And now, perhaps, the universe itself: the ratio of local to CMB-derived expansion rates matching the sixth root of the golden ratio—the same correction that reconciles relativistic and non-relativistic atomic physics—suggests that even the Hubble tension may be a manifestation of the Cantor gate operating at cosmological scale.

8. How to Kill This Theory

Every claim in this paper can be destroyed by data. We welcome the attempt.

Prediction	Confirms if	Kills if	Data source
ρ_6 = Hubble tension	H_0 ratio stays 1.08 ± 0.01	Tension resolves below 2%	DESI + CMB-S4
M-dwarf peak at 1.41	TESS M-dwarf histogram peaks there	Peak is at 1.31 or elsewhere	TESS DR3+
FGK peak at 1.31	Stays dominant in updated catalogs	Peak shifts significantly	Kepler DR25 updates
TRAPPIST d/c = R_base	JWST refines to 1.409 ± 0.002	Shifts outside [1.40, 1.42]	JWST transit timing
Three modes universal	New systems show same 3 values	Different quantization appears	Any new dataset
Exponent m = 1/2	All elements keep Pythagorean form	Any element needs m ≠ 1/2	Atomic spectroscopy

The strongest near-term test is the TESS M-dwarf prediction. If M-dwarf multi-planet systems, analyzed with sufficient statistics, show their dominant spacing at a-ratio ≈ 1.41 (distinct from the FGK peak at ≈ 1.31), the three-mode structure is confirmed across stellar types. If they peak at 1.31 or show no preferred value, the stellar-type dependence predicted here is wrong.

9. Computational Tools and LLM Disclosure

All eigenvalue computations use Python 3 with NumPy, SciPy, and JAX (Metal GPU backend on Apple M4). The Monte Carlo significance tests (100,000 random frameworks × 68 targets × 36,000 combinations) were parallelized on GPU. Formula discovery and adversarial testing were assisted by Claude (Anthropic) and Grok (xAI). All LLM-assisted computations were verified by direct numerical evaluation. The 57 noise targets killed by the Monte Carlo were identified through LLM-mediated statistical analysis; the decision to kill them was made by the author based on the p-values. In accordance with Research Square policy, LLMs are not listed as authors.

10. Data Availability

All code, eigenvalue data, Monte Carlo results, and verification scripts are publicly available at: github.com/thusmann5327/Unified_Theory_Physics. Key files: *algorithms/three_modes_universe.py* (universe-wide scan), *algorithms/theorem_significance.py* (Monte Carlo filter), *algorithms/theorem_recurrence_test.py* (recurrence and Hubble cross-bracket tests), *verification/* (all verification scripts).

Competing Interests

T.A.H. holds 16 provisional patent applications related to engineering applications of the framework. The spectral constants, three-mode values, and cross-scale predictions described in this article are not subject to patent claims and are freely available for academic use under CC BY 4.0.

Acknowledgments

The spectral foundation was built by Damanik, Gorodetski, and Yessen (2016), Maciá (2017), Jagannathan (2021), and Li and Boyle (2023). The Kepler multi-planet period-ratio analyses by He et al. (2019, 2021), Gilbert and Fabrycky (2020), and Steffen and Hwang (2015) provided the FGK population statistics. The TRAPPIST-1 orbital parameters are from Agol et al. (2021). Homonuclear diatomic bond lengths from Huber and Herzberg (1979). Crystal lattice constants from Kittel (2005). Hubble constant values from Planck Collaboration (2020) and Riess et al. (2022). The author thanks Meridian and Ellie May for the long nights, Samuel and Jacob for the right questions, and the Lilliwaup rain for keeping him indoors.

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Lilliwaup, Washington — March 2026

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